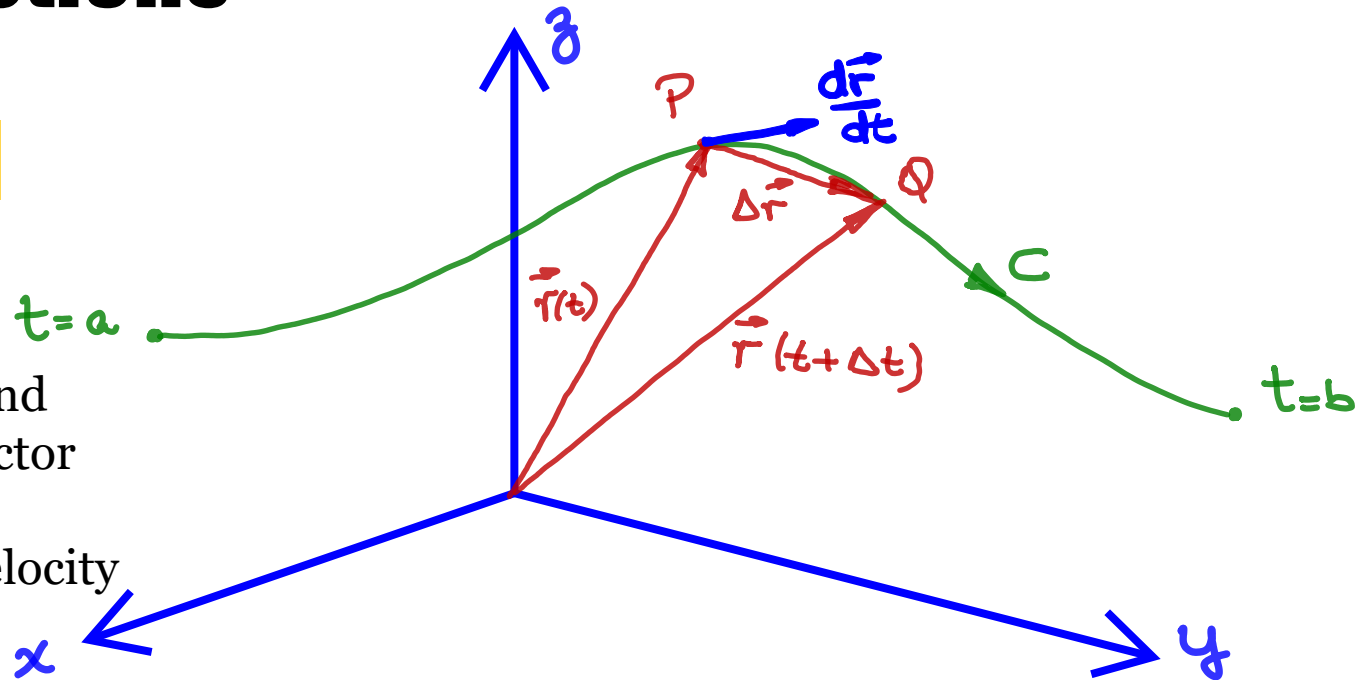


Differentiation and Integration of Vector Functions

Trim Ch 11, S. 11.9, 11.13

Outline:

1. Differentiation and Integration of vector functions
2. Displacement, velocity and acceleration



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Differentiation of Vector Functions

Let us consider the general form of a vector function:

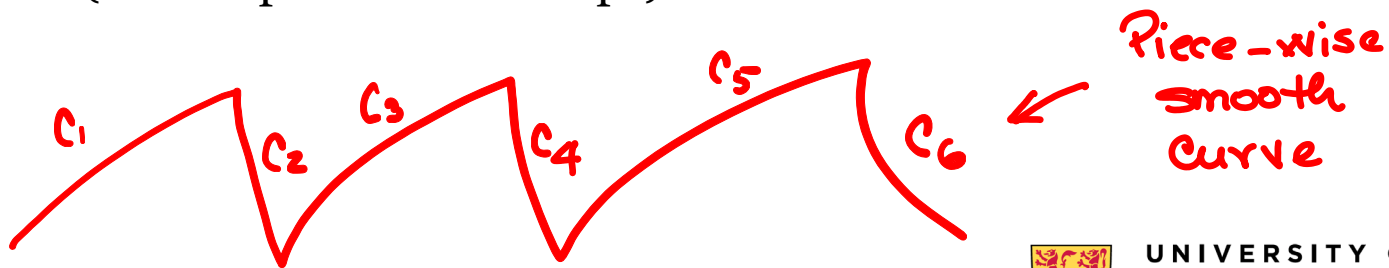
$$\vec{r}(t) = f(t)\hat{i} + g(t)\hat{j} + h(t)\hat{k}, \quad a \leq t \leq b$$

where $\vec{r}(t)$ is differentiable at every point of its domain. The rate of change of $\vec{r}(t)$ as a function of t is given by,

$$\frac{d\vec{r}}{dt} = \vec{r}'(t) = \frac{df(t)}{dt}\hat{i} + \frac{dg(t)}{dt}\hat{j} + \frac{dh(t)}{dt}\hat{k}$$

We can say that,

1. The vector $\frac{d\vec{r}}{dt}$, when different from $\vec{0}$, is the **vector tangent** to the curve C at P.
2. If $\frac{d\vec{r}}{dt} \neq \vec{0}$ and is continuous (i.e. $\frac{d\vec{r}}{dt}$ exists), **C** has a continuously turning tangent and **is smooth** (no sharp corners or cusps).



Problem

Find the parametric equations of the line that is tangent to the smooth curve given by,

$$\vec{r}(t) = \sin t \hat{i} + (t^2 - \cos t)\hat{j} + e^t \hat{k}$$

at $t = 0$.

Solution:

DIRECTION OF TANGENT LINE AT P

$$\vec{v} = \frac{d\vec{r}}{dt} = \cos t \hat{i} + (2t + \sin t) \hat{j} + e^t \hat{k}$$

AT $P_0 \rightarrow t=0$

$$\vec{v}_0(t=0) = \left. \frac{d\vec{r}}{dt} \right|_{t=0} = \hat{i} + \hat{k} = (1, 0, 1)$$

POINT IN THE LINE $\rightarrow P(t=0)$

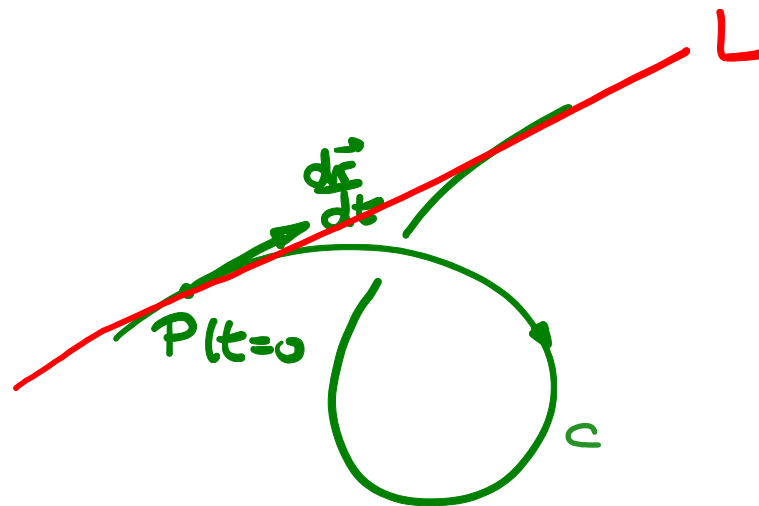


Problem

$$\vec{r}(t=0) \rightarrow P(t=0) \rightarrow (0, -1, 1)$$

PARAMETRIC EQ'S OF LINE TANGENT TO $\vec{r}(t)$ AT $P(t=0)$

$$L \begin{cases} x = 0 + t \\ y = -1 \\ z = 1 + t \end{cases}$$



Orientation of $\frac{d\vec{r}}{dt}$

Case 1: $\Delta t > 0$

$$\Delta\vec{r} = \vec{r}(t+\Delta t) - \vec{r}(t)$$

$$\frac{\Delta\vec{r}}{\Delta t} \rightarrow \frac{d\vec{r}}{dt} \quad \text{POINTS IN THE DIRECTION OF MOTION}$$

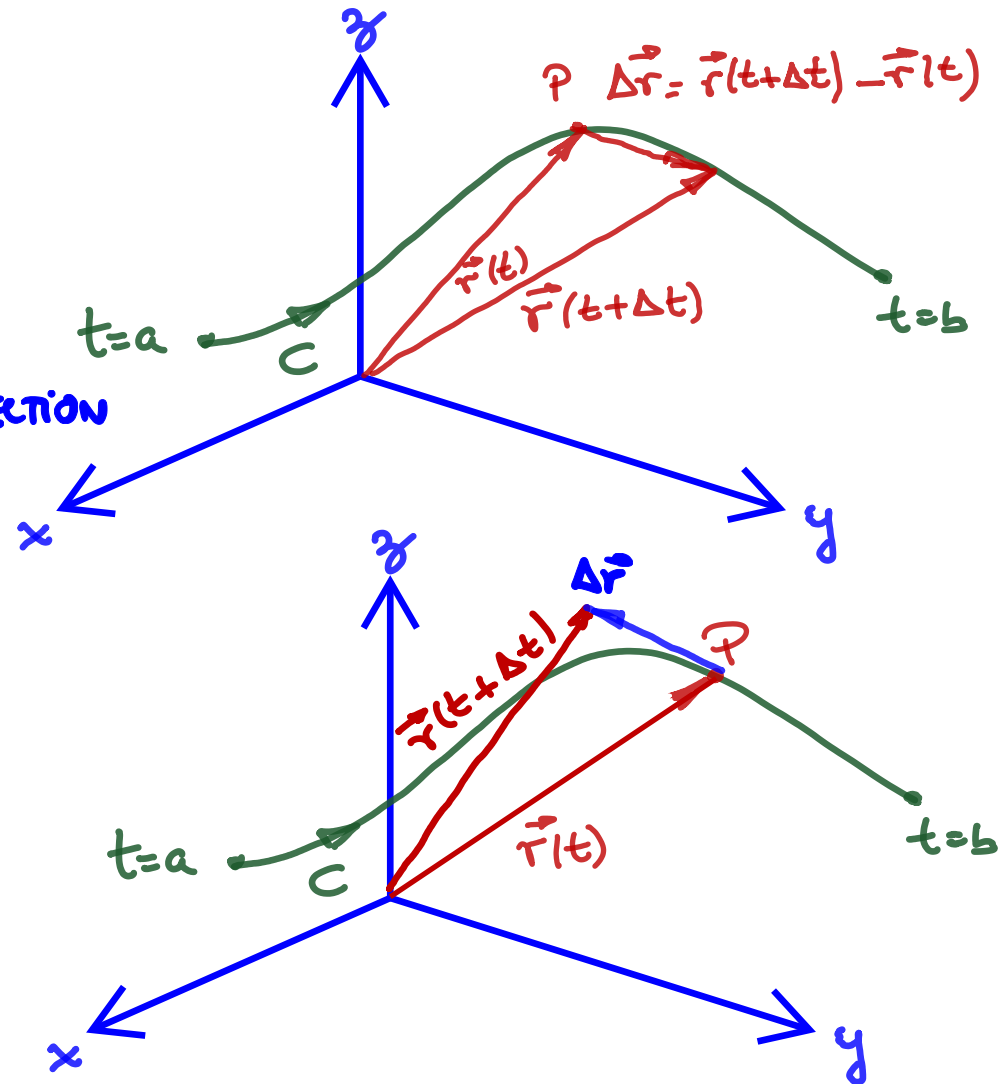
$\Delta t > 0$

Case 2: $\Delta t < 0$

$$\Delta\vec{r} = \vec{r}(t) - \vec{r}(t+\Delta t)$$

$$\frac{\Delta\vec{r}}{\Delta t} \rightarrow \frac{d\vec{r}}{dt} \quad \text{POINTS IN THE DIRECTION OF MOTION}$$

$\Delta t < 0$



Rate of change of $\vec{r}(t)$

For a smooth curve **C** defined by the path:

$$\vec{r}(t) = f(t)\hat{i} + g(t)\hat{j} + h(t)\hat{k}, \quad a \leq t \leq b$$

where f , g , and h are differentiable functions,

$$\frac{d\vec{r}}{dt} = \vec{r}'(t) = \frac{df(t)}{dt}\hat{i} + \frac{dg(t)}{dt}\hat{j} + \frac{dh(t)}{dt}\hat{k}$$

- $\frac{d\vec{r}(t)}{dt}$ always points in the direction in which the parameter “t” increases
- $\frac{d\vec{r}(t)}{dt}$ gives the rate of change of position versus time
- $\frac{d\vec{r}(t)}{dt}$ is never zero; and the particle does not stop or reverse direction

$\frac{d\vec{r}(t)}{dt}$ is just what we need for modeling a particle's velocity!



Velocity, Speed and Acceleration

If $\vec{r}(t)$ is the position vector of a particle moving along a smooth curve C in space; then

$$\vec{v}(t) = \frac{d\vec{r}(t)}{dt}$$

is the particle **velocity vector**, tangent to the curve at any time t and the direction of $\vec{v}(t)$ defines the direction of motion.

Speed is the magnitude of the velocity:

$$|\vec{v}(t)| = \sqrt{\left(\frac{df}{dt}\right)^2 + \left(\frac{dg}{dt}\right)^2 + \left(\frac{dh}{dt}\right)^2}$$

Acceleration is the rate of change of the velocity with respect to time:

$$\vec{a}(t) = \frac{d\vec{v}(t)}{dt} = \frac{d^2\vec{r}(t)}{dt^2}$$

Rules of Differentiation of Vector Functions

Theorem 11.9: Trim Chapter 11, Page 758

If $f(t)$ is a differentiable scalar function of t , and $\vec{u}(t)$ and $\vec{v}(t)$ are differentiable vector-valued functions of t , then,

$$\frac{d}{dt}(f\vec{v}) = \frac{df}{dt}(\vec{v}) + f \frac{d\vec{v}}{dt}$$

$$\frac{d}{dt}(\vec{u} \cdot \vec{v}) = \vec{u} \cdot \frac{d\vec{v}}{dt} + \frac{d\vec{u}}{dt} \cdot \vec{v}$$

$$\frac{d}{dt}(\vec{u} \times \vec{v}) = \vec{u} \times \frac{d\vec{v}}{dt} + \frac{d\vec{u}}{dt} \times \vec{v}$$



Integration of Vector Functions

A vector function $\vec{R}(t)$ is said to be an antiderivative of $\vec{r}(t)$, where $a \leq t \leq b$, if

$$\frac{d\vec{R}(t)}{dt} = \vec{r}(t)$$

The definite integral of $\vec{r}(t)$ is then,

$$\int_a^b \vec{r}(t) dt = \vec{R}(t) + \vec{C}$$

Where $\vec{C}$ is a vector with constant components

Problem

A projectile is fired from the origin over horizontal ground at an initial speed of $500 \frac{m}{s}$ and a launch angle of 60° .

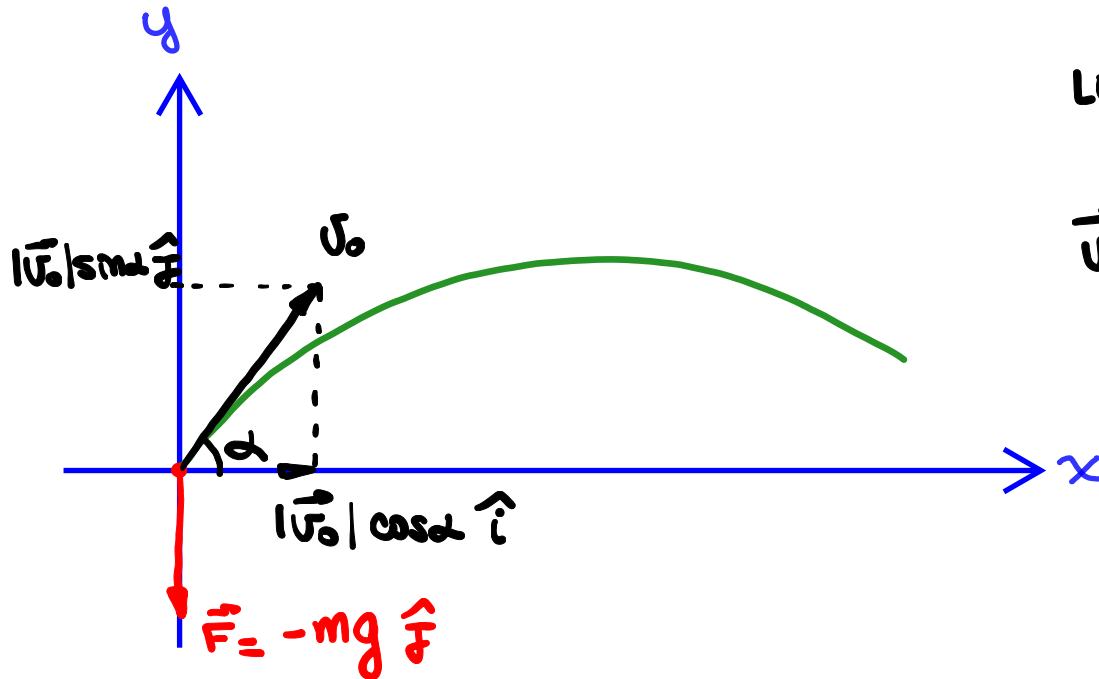
- Where will the projectile be 10 seconds later?
- Find the maximum height the projectile reaches*
- When does the projectile land?*

Answers:

- The projectile will be at $\vec{r}(t) \approx 2500\hat{i} + 3840\hat{j}$
- Maximum height $\approx 9566 \text{ m}$ *
- After $\approx 88.4 \text{ sec}$ *

* Left for your own practice at home

Solution:



LET US CALL $|\vec{v}_0| = v_0$

$$\vec{v}_0 = v_0 \cos \alpha \hat{i} + v_0 \sin \alpha \hat{j} \dots \textcircled{\text{I}}$$

NEWTON'S SECOND LAW ,

$$\vec{F} = m \frac{d^2 \vec{r}}{dt^2} = -mg \hat{j} \rightarrow \text{ODE}$$

$$\frac{d^2 \vec{r}}{dt^2} = -g \hat{j} \dots \textcircled{\text{II}}$$



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INITIAL COND. (I.C.)

$$\vec{r}|_{t=0} = \vec{0} \quad \dots (1)$$

$$\left. \frac{d\vec{r}}{dt} \right|_{t=0} = \vec{v}_0 \quad \dots (2)$$

INTEGRATING BOTH SIDES OF II

$$\underbrace{\frac{d\vec{r}}{dt}}_{\substack{\text{APPLYING} \\ \text{I.C.} \\ \vec{v}_0}} = -\underbrace{(gt)\hat{j}}_{\substack{\vec{v}_0 \\ = 0}} + \vec{C}_1 \quad \rightarrow \quad \vec{C}_1 = \vec{v}_0$$

A SECOND INT. GIVES,

$$\vec{r}(t) = -\frac{1}{2}gt^2\hat{j} + \underbrace{\vec{v}_0 t}_{\substack{\vec{v}_0 \\ = 0}} + \vec{C}_2 \quad \rightarrow \quad \vec{C}_2 = \vec{0}$$

$\underbrace{\vec{r}(0)}_{\substack{\text{APPLYING} \\ \text{I.C.} \\ = \vec{0}}} = \vec{0}$



$$\vec{r}(t) = -\frac{1}{2}gt^2 \hat{j} + \vec{v}_0 t$$

SUBSTITUTING (I),

$$\vec{r}(t) = -\frac{1}{2}gt^2 \hat{j} + v_0 \cos \alpha t \hat{i} + v_0 \sin \alpha t \hat{j}$$

$$\vec{r}(t) = v_0 \cos \alpha t \hat{i} + (v_0 \sin \alpha t - \frac{1}{2}gt^2) \hat{j}$$

FOR OUR PROBLEM $\rightarrow$ POSITION 10 SEC. AFTER FIRING

$$\left. \begin{array}{l} v_0 = 500 \text{ m/s} \\ \alpha = 60^\circ \\ g = 9.8 \text{ m/s}^2 \\ t = 10 \end{array} \right\} \begin{array}{l} \vec{r}(10) = 500\left(\frac{1}{2}\right)(10)\hat{i} + \left(500\left(\frac{\sqrt{3}}{2}\right)10 - \frac{1}{2}(9.8)(100)\right)\hat{j} \\ \vec{r}(10) \approx 2500\hat{i} + 3840\hat{j} \end{array}$$



Summary

- We learned how to differentiate and integrate vector functions
- We also discovered that $\frac{d\vec{r}(t)}{dt}$ is just what we need for modeling a particle's velocity!
 - Points in the direction of motion
 - Gives the rate of change of position vs. time
 - It is never zero (particle never stops or reverse direction)
- The rules of differentiation involving vectors, can also be found in Schaum's, Page 122 (20.23 – 20.27)



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Thank you